

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5.	(a)		[Minimum] energy needed to remove an electron [from the surface or from the metal] Don't accept reference to atoms or ionisation	1			1		
	(b)		Photon energy needed = $3.2 \times 10^{-19} \text{ J} + 1.5 \times 10^{-19} \text{ J}$ [= $4.7 \times 10^{-19} \text{ J}$] or substitution of data into $E_{k \text{ max}} = hf - \phi$ or by implication (1) $f = 7.1 \times 10^{14} \text{ [Hz]}$ (1)		2		2	2	
	(c)	(i)	Photons per second = $\frac{0.3 \text{ [W]}}{4.7 \times 10^{-19} \text{ [J]}}$ ecf (1) = $6.4 \times 10^{17} \text{ [s}^{-1}\text{]}$ Accept $6.5 \times 10^{17} \text{ [s}^{-1}\text{]}$ (1)		2		2	2	
		(ii)	Number of electrons per second = $\frac{0.8 \times 10^{-6} \text{ [A]}}{1.6 \times 10^{-19} \text{ [C]}}$ or by implication Accept slips of powers of 10 (1) = $5.0 \times 10^{12} \text{ [s}^{-1}\text{]}$ (1) Assumes all emitted electrons captured [or equivalent] (1)		3		3	2	
		(iii)	7.8×10^{-6} [or $7.8 \times 10^{-4} \%$] ecf on (c)(i) and (c)(ii) or 8.3×10^{-6} [or $8.3 \times 10^{-4} \%$] if 6×10^{17} used, ecf on (c)(ii)		1		1	1	
			Question 5 total	1	8	0	9	7	0